

Fig_conviction

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Functions

```
#### CONSTANTS ####
wt_fml <- as.formula("~ pop")
model_fml <- as.formula(
  paste0("ratio ~ i(window, region, ref = -1, ref2 = 'EUCA') | region + window")
)

pois_model_fml <- as.formula(
  paste0("n_arrests ~ i(window, region, ref = -1, ref2 = 'EUCA') | region + window")
)
offset_fml <- as.formula("~log_pop")

## Rows: 231336 Columns: 10
## -- Column specification -----
## Delimiter: ","
## chr (4): state, region, method, alt_method
## dbl (6): window, convicted, threat_level, n_arrests, pop_ACS, pop_MPI
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
## `summarise()` has regrouped the output.
## `summarise()` has regrouped the output.
## Fitting model for conv_method: CA_conv

## Wald test, H0: joint nullity of window::-7:region::nonEUCA, window::-6:region::nonEUCA, window::-5:r
## stat = 5.27768, p-value = 2.166e-5, on 6 and 1,400 DoF, VCOV: Custom.[1] "Wald Pre-Trend Test"
## $stat
## [1] 5.277682
##
## $p
## [1] 2.165619e-05
##
## $df1
## [1] 6
##
## $df2
## [1] 1400
##
## $vcov
## [1] "Custom"
```

```

## Warning in .ARP_computeCI(betahat = betahat, sigma = sigma, numPrePeriods =
## numPrePeriods, : CI is open at one of the endpoints; CI length may not be
## accurate
## Warning in .ARP_computeCI(betahat = betahat, sigma = sigma, numPrePeriods =
## numPrePeriods, : CI is open at one of the endpoints; CI length may not be
## accurate

## [1] "M bar Results"
## # A tibble: 8 x 5
##       lb      ub method Delta      Mbar
##   <dbl> <dbl> <chr>  <chr>  <dbl>
## 1  0.192  0.410 C-LF   DeltaRM 0.4
## 2  0.180  0.424 C-LF   DeltaRM 0.5
## 3  0.100  0.517 C-LF   DeltaRM 1.1
## 4  0.0854 0.533 C-LF   DeltaRM 1.2
## 5  0.00740 0.617 C-LF   DeltaRM 1.7
## 6 -0.00874 0.634 C-LF   DeltaRM 1.8
## 7 -0.0894 0.672 C-LF   DeltaRM 2.3
## 8 -0.106 0.672 C-LF   DeltaRM 2.4

## Fitting model for conv_method: CA_nonconv

## Wald test, H0: joint nullity of window::-7:region::nonEUCA, window::-6:region::nonEUCA, window::-5:region::nonEUCA
## stat = 2.29845, p-value = 0.032645, on 6 and 1,400 DoF, VCOV: Custom.[1] "Wald Pre-Trend Test"
## $stat
## [1] 2.298446
##
## $p
## [1] 0.03264526
##
## $df1
## [1] 6
##
## $df2
## [1] 1400
##
## $vcov
## [1] "Custom"
##
## [1] "M bar Results"
## # A tibble: 8 x 5
##       lb      ub method Delta      Mbar
##   <dbl> <dbl> <chr>  <chr>  <dbl>
## 1  0.518  1.84 C-LF   DeltaRM 0.4
## 2  0.494  1.85 C-LF   DeltaRM 0.5
## 3  0.351  2.01 C-LF   DeltaRM 1.1
## 4  0.328  2.03 C-LF   DeltaRM 1.2
## 5  0.185  2.17 C-LF   DeltaRM 1.7
## 6  0.149  2.21 C-LF   DeltaRM 1.8
## 7  0.00596 2.35 C-LF   DeltaRM 2.3
## 8 -0.0298 2.39 C-LF   DeltaRM 2.4

## Fitting model for conv_method: LEA_conv

## Wald test, H0: joint nullity of window::-7:region::nonEUCA, window::-6:region::nonEUCA, window::-5:region::nonEUCA
## stat = 1.90053, p-value = 0.077505, on 6 and 1,400 DoF, VCOV: Custom.[1] "Wald Pre-Trend Test"

```

```

## $stat
## [1] 1.900528
##
## $p
## [1] 0.07750515
##
## $df1
## [1] 6
##
## $df2
## [1] 1400
##
## $vcov
## [1] "Custom"

## Warning in .ARP_computeCI(betahat = betahat, sigma = sigma, numPrePeriods =
## numPrePeriods, : CI is open at one of the endpoints; CI length may not be
## accurate
## Warning in .ARP_computeCI(betahat = betahat, sigma = sigma, numPrePeriods =
## numPrePeriods, : CI is open at one of the endpoints; CI length may not be
## accurate

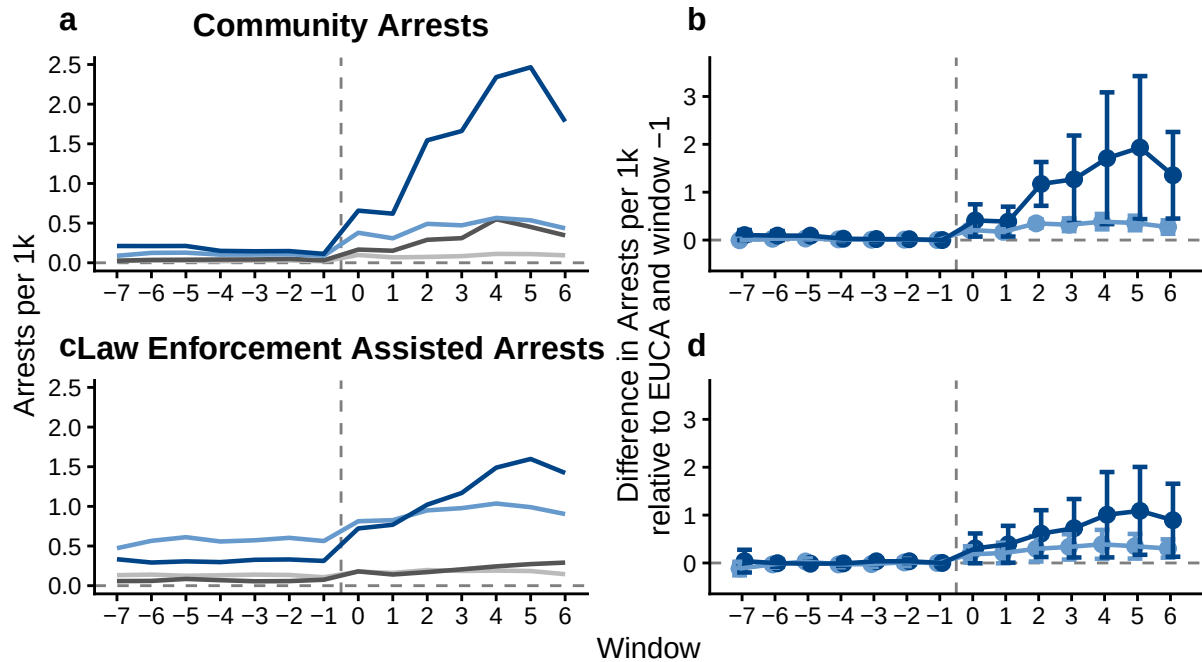
## [1] "M bar Results"
## # A tibble: 8 x 5
##       lb      ub method Delta      Mbar
##   <dbl> <dbl> <chr>  <chr>   <dbl>
## 1  0.0348 0.668 C-LF   DeltaRM 0.4
## 2 -0.0149 0.730 C-LF   DeltaRM 0.5
## 3 -0.386  1.13  C-LF   DeltaRM 1.1
## 4 -0.449  1.19  C-LF   DeltaRM 1.2
## 5 -0.780  1.53  C-LF   DeltaRM 1.7
## 6 -0.846  1.60  C-LF   DeltaRM 1.8
## 7 -1.17   1.65  C-LF   DeltaRM 2.3
## 8 -1.24   1.65  C-LF   DeltaRM 2.4

## Fitting model for conv_method: LEA_nonconv

## Wald test, H0: joint nullity of window::-7:region::nonEUCA, window::-6:region::nonEUCA, window::-5:r
## stat = 1.55846, p-value = 0.1557, on 6 and 1,400 DoF, VCOV: Custom.[1] "Wald Pre-Trend Test"
## $stat
## [1] 1.558461
##
## $p
## [1] 0.1556995
##
## $df1
## [1] 6
##
## $df2
## [1] 1400
##
## $vcov
## [1] "Custom"
##
## [1] "M bar Results"
## # A tibble: 8 x 5

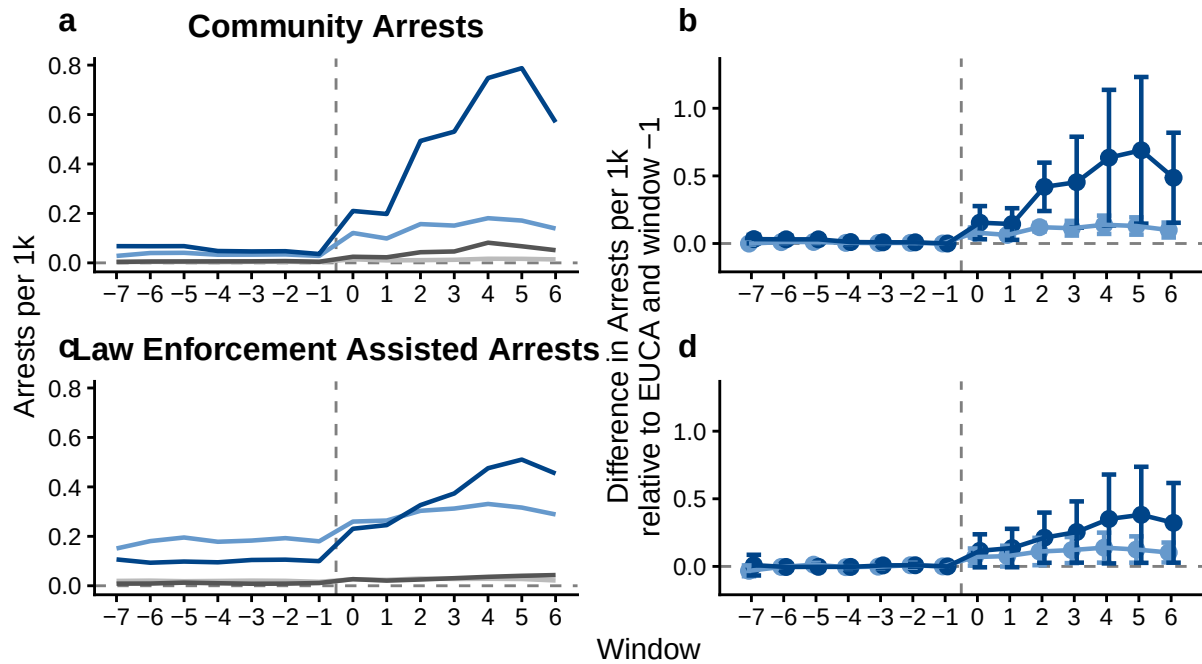
```

##	lb	ub	method	Delta	Mbar
##	<dbl>	<dbl>	<chr>	<chr>	<dbl>
## 1	0.240	1.32	C-LF	DeltaRM	0.4
## 2	0.221	1.36	C-LF	DeltaRM	0.5
## 3	0.0141	1.67	C-LF	DeltaRM	1.1
## 4	-0.0423	1.73	C-LF	DeltaRM	1.2
## 5	-0.315	2.03	C-LF	DeltaRM	1.7
## 6	-0.381	2.09	C-LF	DeltaRM	1.8
## 7	-0.682	2.40	C-LF	DeltaRM	2.3
## 8	-0.738	2.47	C-LF	DeltaRM	2.4



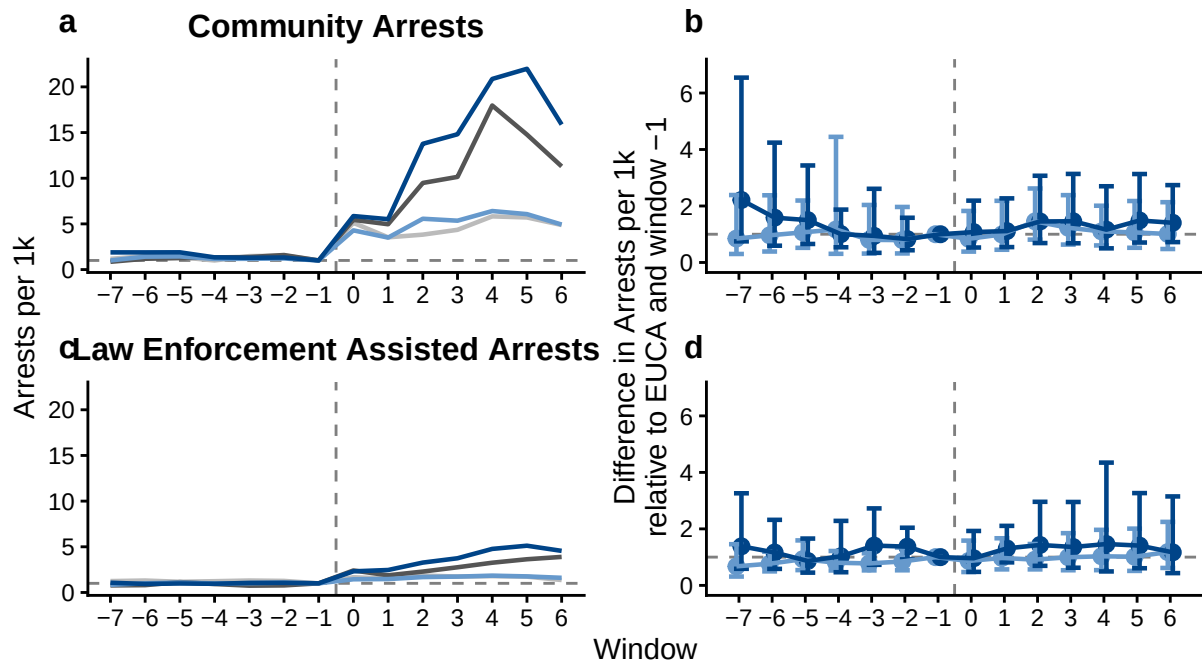
n & Status — EUCA, Non-Convicted — EUCA, Convicted — Non-EUCA, Non-Convicted — Non-EUCA, Convicted

```
## `summarise()` has regrouped the output.
## `summarise()` has regrouped the output.
## Fitting model for conv_method: CA_conv
## Fitting model for conv_method: CA_nonconv
## Fitting model for conv_method: LEA_conv
## Fitting model for conv_method: LEA_nonconv
## i Summaries were computed grouped by state, region, window, convicted,
##   threat_level, method, alt_method, and conv_method.
## i Output is grouped by state, region, window, convicted, threat_level, method,
##   and alt_method.
## i Use `summarise(.groups = "drop_last")` to silence this message.
## i Use `summarise(.by = c(state, region, window, convicted, threat_level,
##   method, alt_method, conv_method))` for per-operation grouping
## (`?dplyr::dplyr_by`) instead.
```



1 & Status — EUCA, Non-Convicted — EUCA, Convicted — Non-EUCA, Non-Convicted — Non-EUCA, Convicted

```
## Fitting model for conv_method: CA_conv
## Fitting model for conv_method: CA_nonconv
## Fitting model for conv_method: LEA_conv
## Fitting model for conv_method: LEA_nonconv
```



1 & Status — EUCA, Non-Convicted — EUCA, Convicted — Non-EUCA, Non-Convicted — Non-EUCA, Convicted

Test for significance

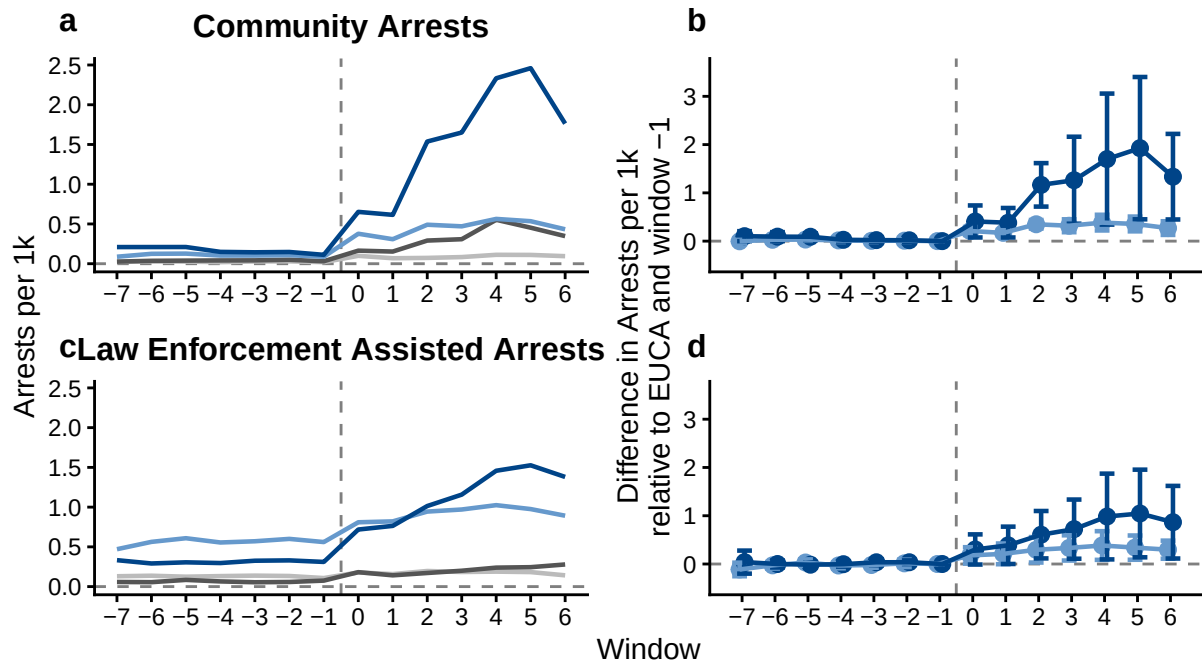
```

## Rows: 231336 Columns: 10
## -- Column specification -----
## Delimiter: ","
## chr (4): state, region, method, alt_method
## dbl (6): window, convicted, threat_level, n_arrests, pop_ACS, pop_MPI
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
## `summarise()` has regrouped the output.
## `summarise()` has regrouped the output.

## [1] "LEA conv:"
## [1] 0.002757093
## [1] 4.798367 1.656729 7.940005
## [1] "LEA nonconv:"
## [1] 0.004564873
## [1] 6.449496 1.992528 10.906464
## [1] "CA conv:"
## [1] 4.615436e-16
## [1] 2.478110 1.880035 3.076186
## [1] "CA nonconv:"
## [1] 6.202505e-05
## [1] 8.725421 4.455272 12.995571

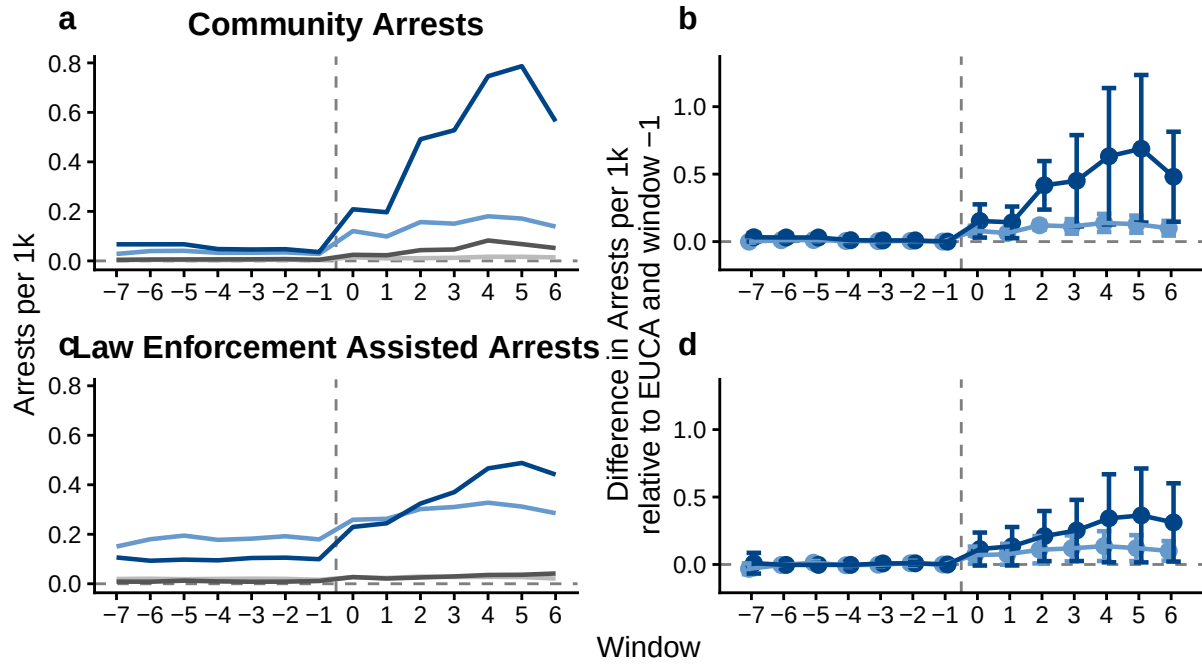
## Rows: 231336 Columns: 10
## -- Column specification -----
## Delimiter: ","
## chr (4): state, region, method, alt_method
## dbl (6): window, convicted, threat_level, n_arrests, pop_ACS, pop_MPI
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
## `summarise()` has regrouped the output.
## `summarise()` has regrouped the output.
## Fitting model for conv_method: CA_conv
##
## Fitting model for conv_method: CA_nonconv
##
## Fitting model for conv_method: LEA_conv
##
## Fitting model for conv_method: LEA_nonconv

```



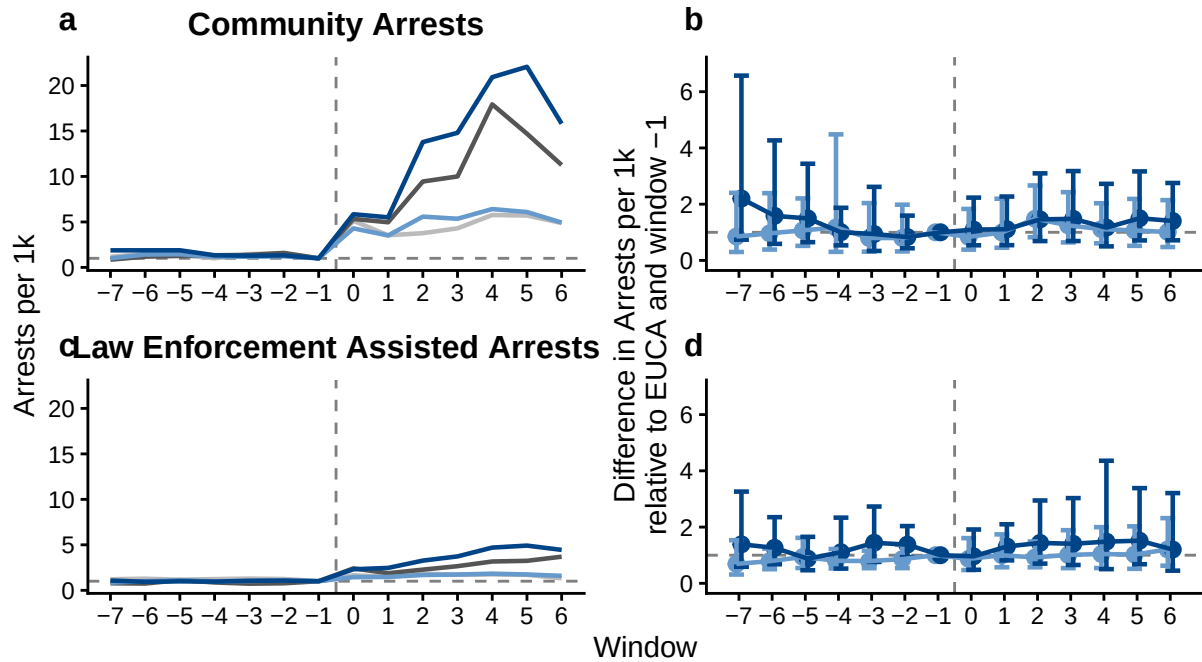
n & Status — EUCA, Non-Convicted — EUCA, Convicted — Non-EUCA, Non-Convicted — Non-EL

```
## `summarise()` has regrouped the output.
## `summarise()` has regrouped the output.
## Fitting model for conv_method: CA_conv
## Fitting model for conv_method: CA_nonconv
## Fitting model for conv_method: LEA_conv
## Fitting model for conv_method: LEA_nonconv
## i Summaries were computed grouped by state, region, window, convicted,
##   threat_level, method, alt_method, and conv_method.
## i Output is grouped by state, region, window, convicted, threat_level, method,
##   and alt_method.
## i Use `summarise(.groups = "drop_last")` to silence this message.
## i Use `summarise(.by = c(state, region, window, convicted, threat_level,
##   method, alt_method, conv_method))` for per-operation grouping
##   (`?dplyr::dplyr_by`) instead.
```



1 & Status — EUCA, Non-Convicted — EUCA, Convicted — Non-EUCA, Non-Convicted — Non-EUCA, Convicted

```
## Fitting model for conv_method: CA_conv
## Fitting model for conv_method: CA_nonconv
## Fitting model for conv_method: LEA_conv
## Fitting model for conv_method: LEA_nonconv
```



1 & Status — EUCA, Non-Convicted — EUCA, Convicted — Non-EUCA, Non-Convicted — Non-EUCA, Convicted